MASON DUBOEF

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Website · LinkedIn · GitHub

EDUCATION

University of Massachusets Amherst *Dec 2026 (expected)*
M.S. in Computer Science
GPA: 3.94/4.0
Relevant courses: Reinforcement Learning | AI Alignment | Machine Learning | Artificial Intelligence | Algorithms, Game Theory & Fairness | Cyber Effects | Linear Algebra

Rensselaer Polytechnic Institute (RPI) *June 2023*
B.S. in Computer Science (Concentration in AI & Data)
Relevant courses: Intro to AI (later mentored) | Economics & Computation | Intro to Network Science | Foundations of CS (Theory) | Intro to Algorithms | Data Structures | Multivariable Calculus & Matrix Algebra | Differential Equations | Operating Systems | Prog. Languages | Computability & Logic

RESEARCH INTERESTS

I am broadly interested in reinforcement learning, particularly agents that learn through environment interaction with objective structures that richly express human preferences (MORL, IRL, alignment).

RESEARCH EXPERIENCE

Contrastive Preference Learning under Stochastic Dominance *Feb 2026 - Present*
Supervisor: Prof. Scott Neikum SCALAR Lab - UMass Amherst

- Extending contrastive preference learning, an algorithm for learning optimal policies from preferences reported over trajectories, without the need for a reward function/RL framework
- New learning objective, enforcing first-order stochastic dominance via optimal transport

Algorithmic Fair Allocation for Food Rescue *Sep 2025 - Dec 2025*
Supervisor: Prof. Yair Zick FED Lab - UMass Amherst

- Integer linear program optimized routing of drivers from food donors to receiving agencies
- Produced fair and efficient allocation of food under changing dynamics and stochastic availability

Interpretable Prediction & Large-Scale Analysis of Judging in Boxing *Jan 2024 - Present*
Supervisor: Dr. Allan Svejstrup Nielsen Jabbr

- Developed autonomous judging system for boxing with accuracy within the range of top-level judges
- Supervised learning mapped stats output by computer vision system onto judges' scores
- Evaluation of top judges and analysis of their stylistic differences
- Oral presentation at *2026 MIT Sloan Sports Analytics Conference* as a competition finalist

Stable Matching in OPRA Voting Platform *Sep 2022 - Jun 2023*
Supervisor: Prof. Lirong Xia RPI

- Back-end Django development on OPRA, an online preference reporting and aggregation system
- Added support for stable matching problems, including deployment of different matching algorithms

PUBLICATION

duBoef, M., Romeas, T., Charbonneau, M., & Nielsen, A. S. Interpretable Prediction and Large-Scale Analysis of Judging in Professional Boxing. *(Finalist, 7 of 197 submissions selected) MIT Sloan Sports Analytics Conference 2026.*

PROFESSIONAL EXPERIENCE

Jabbr - Research Intern *Jan 2024 - Present*
Reference: Dr. Allan Svejstrup Nielsen (CEO)

- ML-based research on judging in professional boxing
- See "Research Experience" section for details

Mammoth Media - Data Science Intern *Aug 2021 - Jan 2022*
References: Solene Schwartz (COO) and Zachary Chow (General Manager)

- Statistical analysis of TikTok ads, informing branded content creation & media buying strategy
- Automated production and delivery of performance dashboards for clients
- Media buying, personally managed 5-6 figure monthly spend for several notable brands

Meta - Data Challenge Finalist *Apr 2021 - Aug 2021*

- Four-month training program at Meta, mentored by Facebook data scientists and engineers
- SQL courses, final project on market viability of telenovelas for OTT streaming services

Luum.io - Software Engineering Intern *May 2020 - Aug 2020*
Reference: Sebastien Gouin-Davis (CEO)

- Android app design, testing and development for lighting control platform
- Created PHP-based cost estimation tool for online marketing and customer acquisition

PERSONAL PROJECTS

NYC Subway Challenge *Oct 2025 – Dec 2025*

- Hierarchical reinforcement learning system to find a route for the NYC Subway Challenge (a minimum spanning walk problem) using value iteration and between-ness clustering

Willow *Mar 2023 - Jun 2023*

- Working under Prof. Bram Van Heuveln (RPI)
- Expanded web app used to build and assign truth trees, enabling Davis-Putnam type logic problems

Have I Been Gerrymandered? *Sep 2022 – Dec 2022*

- An interactive online map indicating how gerrymandered individual congressional districts are
- Developed novel extension to efficiency gap, a measure of district fairness given electoral data

Dynamic Subway Tolling for Congestion Deterrence *Mar 2022 - Jun 2022*

- Devised optimal toll pricing to deter congestion and promote efficiency on NYC's 1 Line
- Used Nash equilibrium analysis and traffic simulation based on MTA data

Automatic Door Control *Jun 2021 - Aug 2021*

- Android development and circuit design for accessibility project, enabling remote opening of doors

SKILLS

Languages	Python, C/C++, Java, Django, SQL, TypeScript, Kotlin, React.js, Prolog, Lisp, Haskell, & MIPS Assembly
Tools	TensorFlow, PyTorch, Sklearn, Pandas, Numpy, Git, Matplotlib, Jupyter Notebook, Gurobi, Postman, Gephi, LaTeX, Beautiful Soup & WordPress